

Spin geometry

Problem sheet 2

Problem 1. *Ideas from Atiyah–Bott–Shapiro*

- (1) Suppose that $M = M^0 \oplus M^1$ is a graded Clifford module. Show that there is an isomorphism of Clifford modules

$$(1) \quad M \cong \mathbb{C}\ell_n \otimes_{\mathbb{C}\ell_n^0} M^0.$$

- (2) Let $\mathcal{M}_n^{\mathbb{C}}$ denote the Grothendieck group of graded modules for the complex Clifford algebra $\mathbb{C}\ell_n$ and let $\mathcal{A}_\bullet^{\mathbb{C}}$ be the cokernel of the homomorphism $\mathcal{M}_{n+1}^{\mathbb{C}} \xrightarrow{i^*} \mathcal{M}_n^{\mathbb{C}}$ induced from the inclusion $i: \mathbb{C}\ell_n^{\mathbb{C}} \rightarrow \mathbb{C}\ell_{n+1}^{\mathbb{C}}$. In class, we showed that $\mathcal{A}_\bullet^{\mathbb{C}}$ is a $\mathbb{Z}$ -graded ring. Show that there is an isomorphism of graded rings $\mathcal{A}_\bullet^{\mathbb{C}} \simeq \mathbb{Z}[\alpha]$ where α is of degree two.
- (3) For $Y \subset X$ a closed subspace of a topological space X we have defined the groups $L_n(X, Y)$ in class. Show that the natural map $L_n(X, Y) \rightarrow L_{n+1}(X, Y)$ is an isomorphism.

Problem 2. *The second Steifel–Whitney class*

In this problem you will show that a manifold M admits a spin structure if and only if $w_2(M) = 0$.

- (1) Fix a metric and an orientation on M . Pick a good cover $\mathcal{U} = \{U_\alpha\}$ of M which trivializes the frame bundle Fr_M^{SO} and suppose $\{g_{\alpha\beta}: U_{\alpha\beta} \rightarrow SO(n)\}$ are the transition functions. Argue that there exists lifts $\{\tilde{g}_{\alpha\beta}: U_{\alpha\beta} \rightarrow Spin(n)\}$ such that $\tilde{g}_{\alpha\alpha} = \mathbb{1}$ and $\tilde{g}_{\alpha\beta} = g_{\beta\alpha}^{-1}$.
- (2) Define $\epsilon_{\alpha\beta\gamma} = \tilde{g}_{\alpha\gamma}\tilde{g}_{\beta\gamma}\tilde{g}_{\gamma\beta}$. Show that $\epsilon_{\alpha\beta\gamma}(p) \in \ker(Spin(n) \rightarrow SO(n)) = \mathbf{Z}/2$.
- (3) Check that $\{\epsilon_{\alpha\beta\gamma}\}$ defines a Čech cocycle.
- (4) Show that this cocycle is independent of the choice of a lift in (a). Thus, $\{\epsilon_{\alpha\beta\gamma}\}$ gives a well-defined element

$$w_2(M) \stackrel{\text{def}}{=} [\{\epsilon_{\alpha\beta\gamma}\}] \in \check{H}^2(M; \mathbf{Z}/2) \simeq H^2(M; \mathbf{Z}/2)$$

which vanishes if and only if M admits a spin structure.

Problem 3. *Spin structures on Kahler manifolds*

- (1) Recall what an almost complex manifold, what a complex manifold is, and what the canonical line bundle is.
- (2) Endowing $\mathbf{R}^{2n}$ with its standard almost complex structure defines a homomorphism $i: U(n) \rightarrow SO(2n)$. Show that the induced map

$$(2) \quad i^*: H^1(SO(2n); \mathbf{Z}/2) \rightarrow H^1(U(n); \mathbf{Z}/2)$$
 is injective.
- (3) Consider the determinant map $\det: U(n) \rightarrow U(1)$. Show that

$$(3) \quad \det^*: H^1(U(1); \mathbf{Z}/2) \rightarrow H^1(U(n); \mathbf{Z}/2)$$
 is injective.
- (4) Argue that spin structures on an almost complex manifold X are in bijective correspondence with double coverings of the $U(1)$ bundle $\det(Fr_X^{SO(2n)})$ which restrict to the squaring map $U(1) \rightarrow U(1), z \mapsto z^2$ on each fiber.
- (5) Further, show that on a complex manifold X spin structures are in bijective correspondence with the set of isomorphism classes of holomorphic line bundles L on X which satisfy $L^{\otimes 2} \simeq K_X$. Such line bundles L are called “square-roots” of the canonical bundle.

Let G, H be connected Lie groups and assume that G is simply connected. If we have a short exact sequence

$$(4) \quad 1 \rightarrow K \rightarrow G \rightarrow H \rightarrow 1$$

with K discrete, then $\pi_1(H) \simeq K$.

Let's apply this to the short exact sequence

$$(5) \quad 1 \rightarrow \mathbf{Z} \rightarrow SU(n) \times \mathbf{R} \rightarrow U(n) \rightarrow 1$$

where the surjective map is $(A, t) \mapsto e^{it}A$. Since $SU(n)$ is simply connected we see that $\pi_1(U(n)) \simeq \mathbf{Z}$.

Now, consider the fiber bundle

$$(6) \quad SU(n) \rightarrow U(n) \xrightarrow{\det} S^1.$$

Again, since $SU(n)$ is simply connected we see that the induced map $\mathbf{Z} \rightarrow \pi_1(U(n)) \rightarrow \pi_1(S^1) = \mathbf{Z}$ is an isomorphism. By the universal coefficient theorem, it follows that the map $\det^*: H^1(S^1; \mathbf{Z}) \rightarrow H^1(U(n); \mathbf{Z})$ is an isomorphism.